

101991241 | RS485 Moisture & EC & Temperature sensor for Soil, Substrate, Rockwool, Cocopeat

SKU: 101991241

Product Overview

An IP68 RS485 substrate moisture, EC, and temperature sensor designed to monitor volumetric water content (VWC), temperature, and electrical conductivity in soil and soilless substrates. It measures VWC from 0-100%, EC from 0-20,000 uS/cm, and temperature from -40degC to +80degC. Suitable for use with cocopeat, rockwool, potting soil, and similar substrates.

Features

Suitable for Soil, Rockwool, Cocopeat and Substrate: Designed for both mineral soils and soilless substrates like rockwool, potting soil, and coco peat, this sensor offers precise, reliable measurements essential for optimizing plant growth.

High Precision and Sensitivity: The sensor delivers high precision and stable performance in various soil conditions, with stainless steel probes designed to optimize EC measurements and minimize substrate disturbance.

Durable Design for Long-Term Use: This IP68-rated sensor is built for tough conditions, suitable for long-term monitoring in applications like scientific research, water-saving irrigation, horticulture, and precision agriculture.

Easy Integration and Versatile Application: This sensor can be seamlessly integrated with the SenseCAP S2100 LoRaWAN datalogger for efficient data collection and management. Monitor the key indicators of your soil's health with our advanced IP68 RS485 Soil Moisture, Temperature, and Electrical Conductivity (EC) Sensor, designed for both mineral soils and substrates like rockwool, potting soil, and coco peat, offering precise measurements essential for optimizing plant growth by integrating VWC readings derived from measuring the dielectric constant, and providing the accuracy needed for greenhouses, outdoor farms, and environmental monitoring. We offer the SenseCAP S2100 - LoRaWAN® Data Logger and SenseCAP Sensor Hub 4G Data Logger suitable for RS485 sensors, you can transmit the sensor data to public LoRaWAN® network servers(Helium, TTN, etc.,) or MQTT private servers.

Product Image



Product image

Specifications

Output Interface: RS485, Modbus-RTU

Power Supply: 3.9-28V/DC

Power Consumption: RS485: Quiescent Current : <4mA Measuring Current : <20mA during 150ms measurement

Substrate Moisture Measurement: Apparent Dielectric Permittivity: Range:1-81 (air - water);Resolution: 0.1(1-40), 0.5(40.00-81);Accuracy: +/-1(1-40), +/-10% of Readings(40-81);Volumetric Water Content (VWC):Range: 0-100%;Resolution:

0.1%;Accuracy: +/-3%;The VWC range is dependent on the media the sensor is calibrated to. A default calibration will accommodate the necessary ranges for most substrates.

Substrate EC Measurement: Range: 0-20ds/m; Resolution: 0.01ds/m; Accuracy: +/-5%(0-7ds/m); +/-10% of Readings(7-20ds/m)Temperature compensation: 0-50degC

Substrate Temperature Measurement: Range: -40-80degC;Resolution: 0.1degC;Accuracy:+/-0.5degC;

IP Ratings: IP68

Operating Temperature: -40-80degC

Sensor Sealed: Epoxy resin

Installation: Surface or buried installation

Cable Length: 5 meters

Dimension: 90x55x75mm

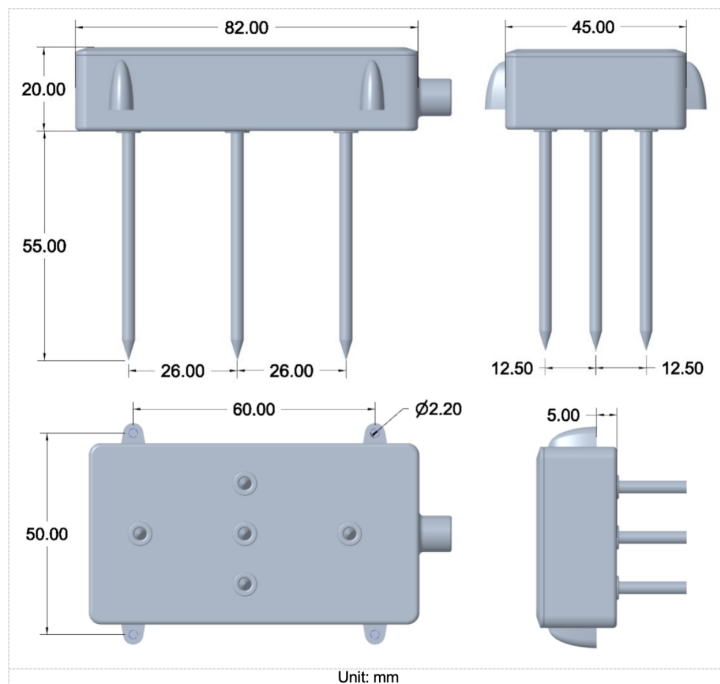
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Hardware Overview



Hardware overview image

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